Work Ethics Application Setup Guide

Full setup guide for Docker and manual local development.

Backend: Go + PostgreSQL | Frontend: Next.js + TypeScript | Date: 2026-03-20

1. Prerequisites

Docker Setup

- Docker Desktop (or Docker Engine + Docker Compose)

Manual Setup

- Go 1.25.x
- Node.js LTS + npm
- PostgreSQL

2. Environment Variables

2.1 Root .env (for main docker compose)

```
POSTGRES_USER=work_ethics_user POSTGRES_PASSWORD=work_ethics_password
POSTGRES_DB=work_ethics_db POSTGRES_HOST_AUTH_METHOD=scram-sha-256
DB_PORT=5433 APP_PORT=8080 FRONTEND_PORT=3000
JWT_SECRET=replace_this_with_a_secure_value
NEXT_PUBLIC_API_URL=http://localhost:8080/api/v1
ADMIN_FULL_NAME=System Admin ADMIN_EMAIL=admin@work-ethics.local
ADMIN_PASSWORD=change_this_admin_password
API_KEY=replace_with_strong_api_key
NEXT_PUBLIC_API_KEY=replace_with_strong_api_key
```

2.2 Backend .env (manual)

```
APP_PORT=8080 DB_HOST=localhost POSTGRES_USER=postgres
POSTGRES_PASSWORD=password POSTGRES_DB=backend_db DB_PORT=5432
DB_SSLMODE=disable JWT_SECRET=supersecretkey ADMIN_FULL_NAME=System
```

```
Admin ADMIN_EMAIL=admin@work-ethics.local ADMIN_PASSWORD=Admin@123456
API_KEY=replace_with_strong_api_key
```

2.3 Frontend .env.local (manual)

```
NEXT_PUBLIC_API_URL=http://localhost:8080/api/v1
NEXT_PUBLIC_API_KEY=replace_with_strong_api_key
```

Keep API_KEY and NEXT_PUBLIC_API_KEY the same to support login routes.

3. Setup Option A: Docker (Recommended)

1. Create root .env in the project root.
2. Run all services from project root:

```
docker compose up --build -d
```

Useful Docker commands

```
docker compose ps docker compose logs -f docker compose down docker
compose down -v
```

4. Setup Option B: Manual Local Run

4.1 Start PostgreSQL

Make sure PostgreSQL is running and matches backend .env values.

4.2 Start backend

```
cd backend go mod tidy go run ./cmd/server
```

4.3 Start frontend

```
cd frontend npm install npm run dev
```

5. Access URLs

- Frontend: http://localhost:3000
- Backend base: http://localhost:8080
- API health: http://localhost:8080/api/v1/health

6. Troubleshooting

- Login/API key issues: verify API_KEY and NEXT_PUBLIC_API_KEY are set and identical.
- Frontend cannot reach API: verify NEXT_PUBLIC_API_URL points to http://localhost:8080/api/v1.
- DB errors: verify host, port, user, password, and database values in env files.

7. Demo Video

Demo URL: PASTE_DEMO_VIDEO_URL_HERE